EQUATION OF SINUSOIDAL FUNCTION

PLANE TRIGONOMETRY

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Equation of Sinusoidal Form

$$y = a\sin(bx + c) + d$$

Amplitude

Midline

Phase Shift

Period

EQUATION OF A SINUSOIDAL FORM

$$y = a\sin(bx + c) + d$$

where,

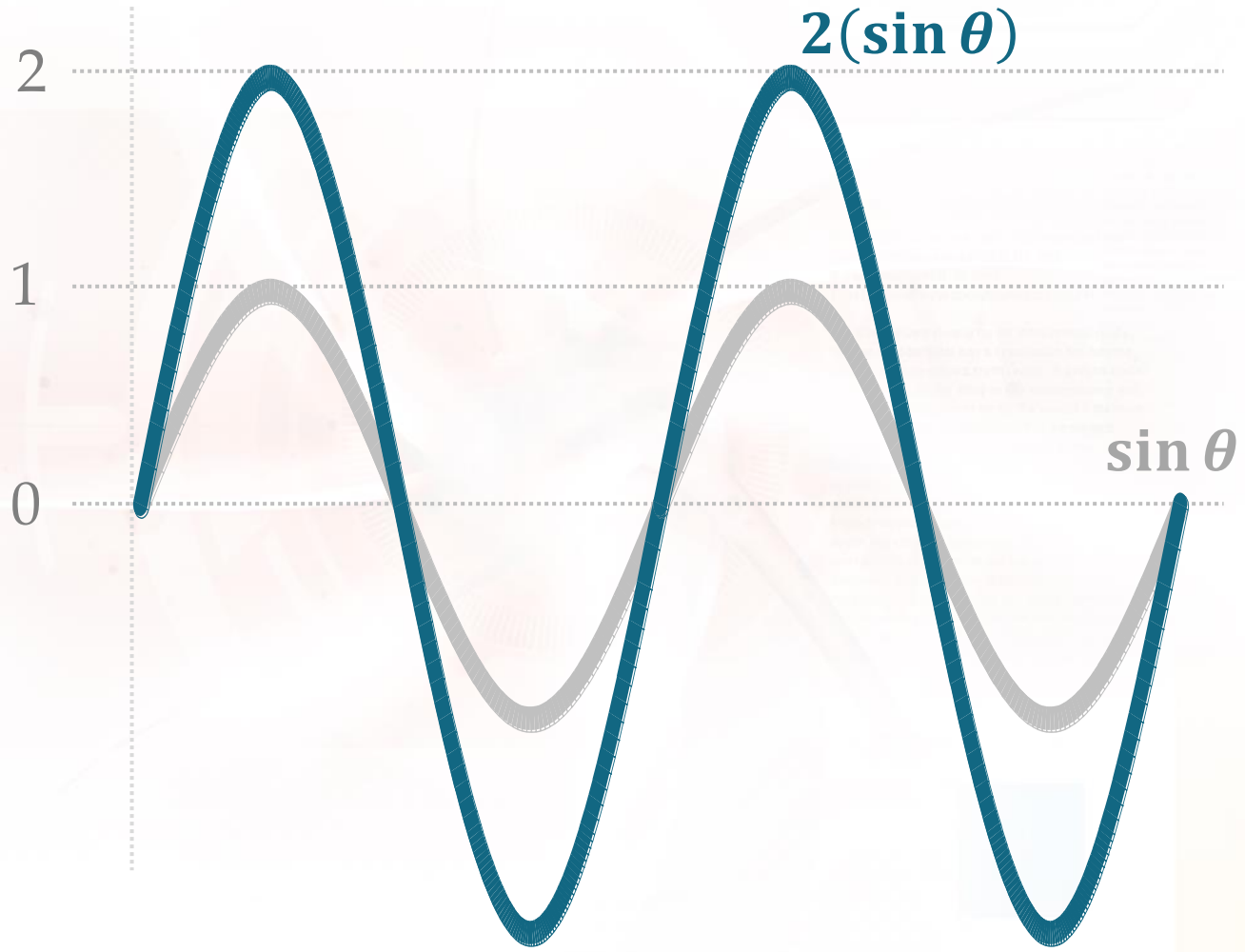
a = amplitude

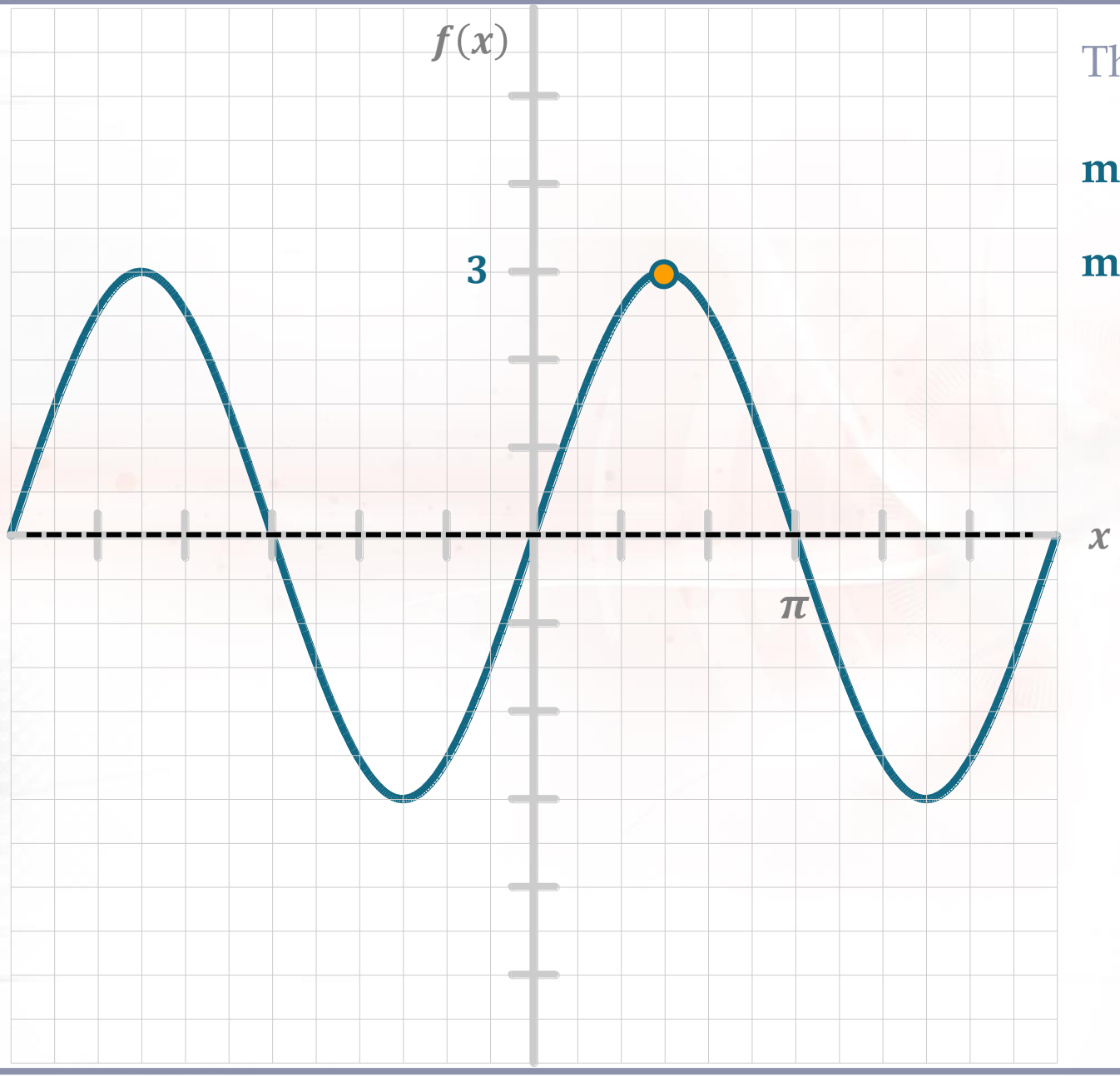
c = phase shift

d = midline

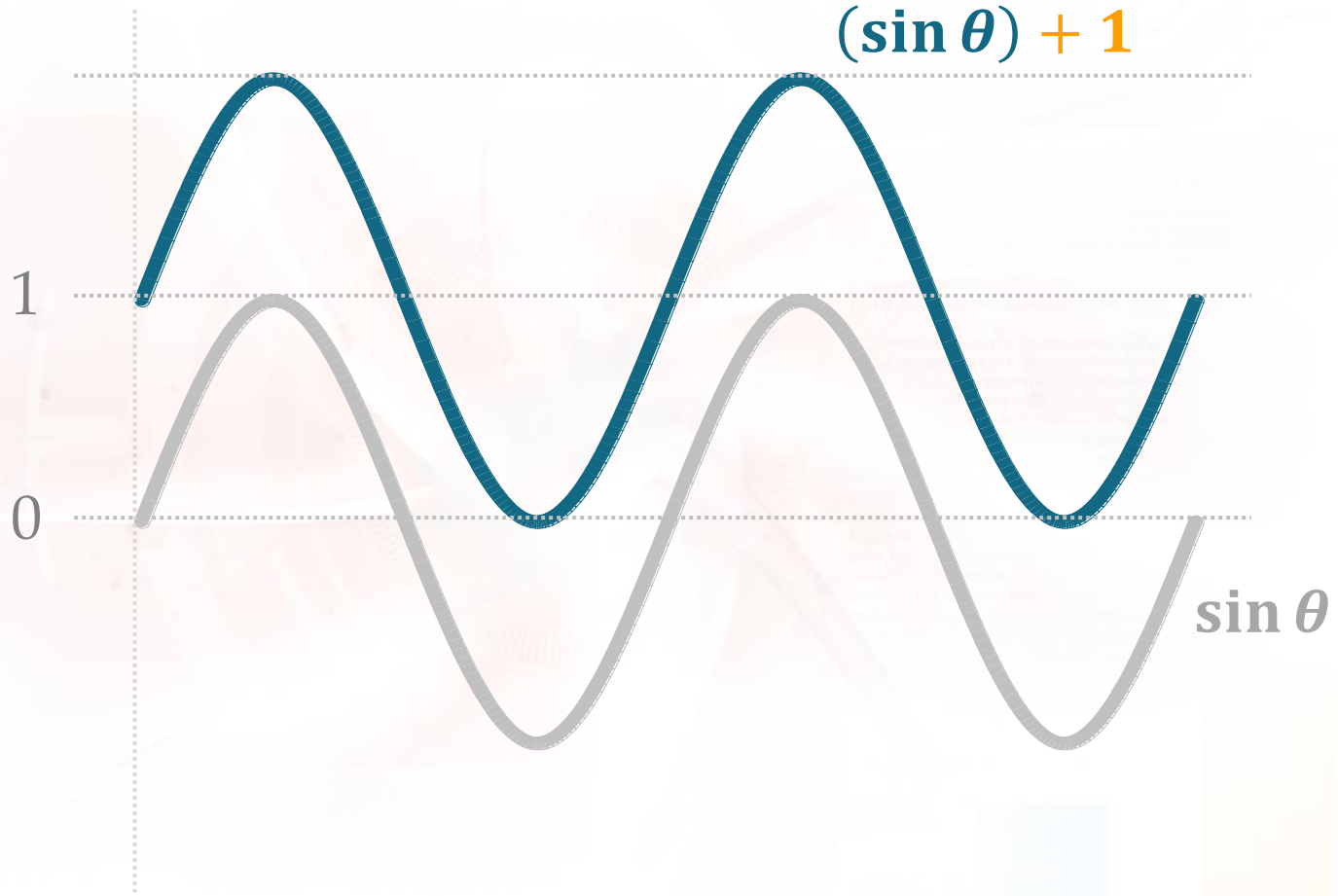
period $= \frac{2\pi}{b}$

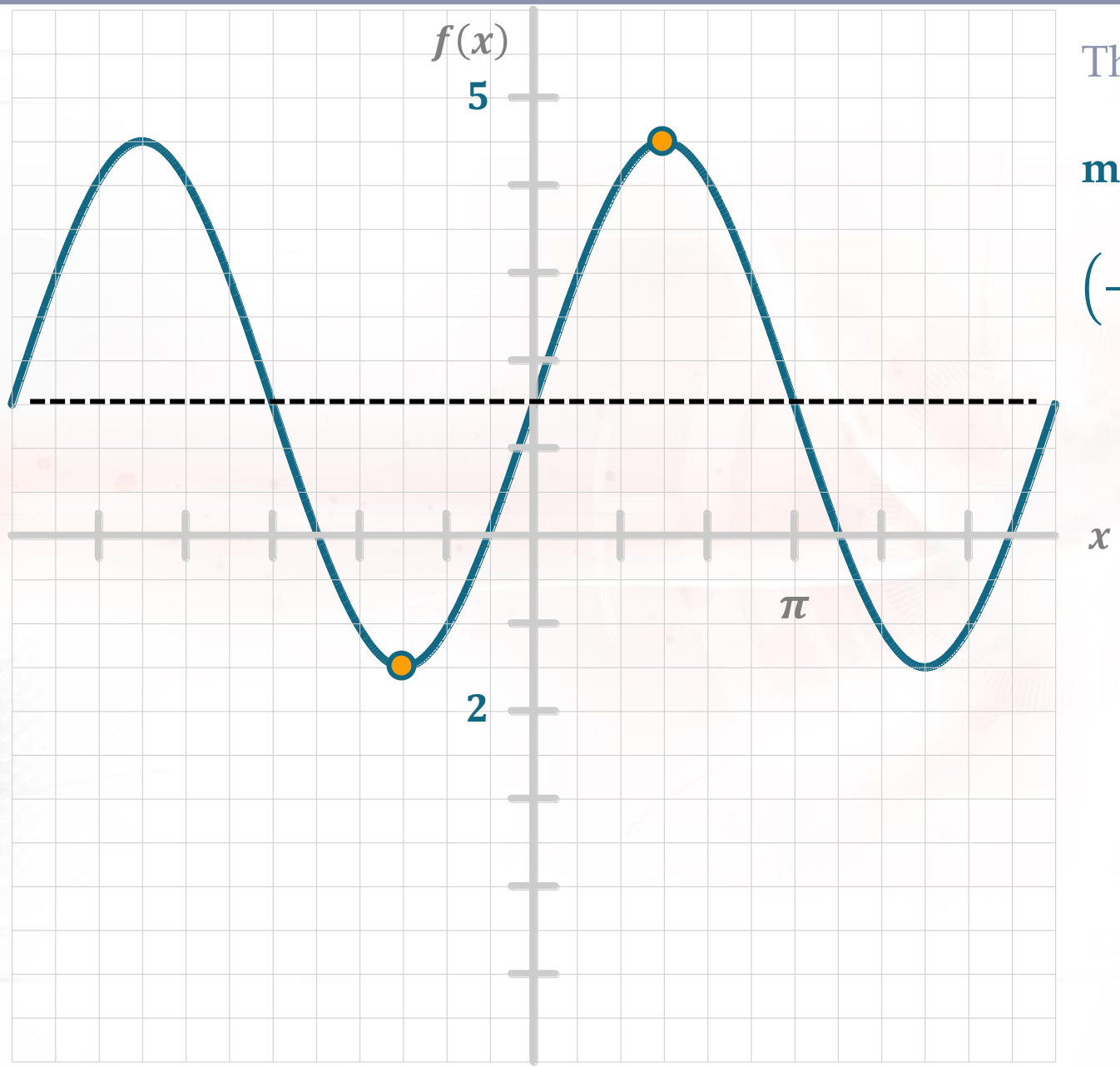
AMPLITUDE





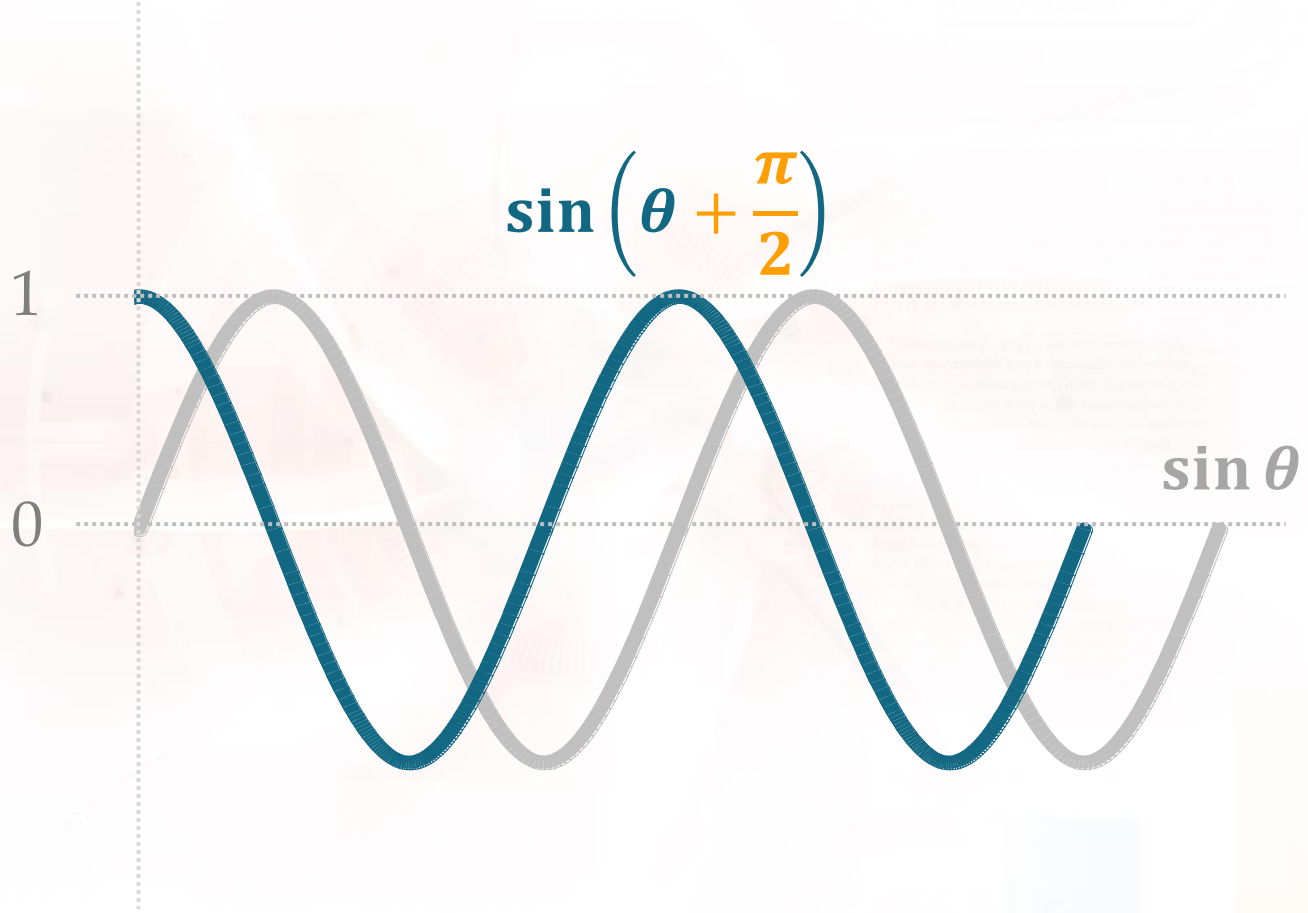
The given sinusoidal function graph has its **maximum** point at $\left(\frac{\pi}{2}, 3\right)$ and it intersects its **midline** at $(0, 0)$. Find the **formula** for **f(x)**.

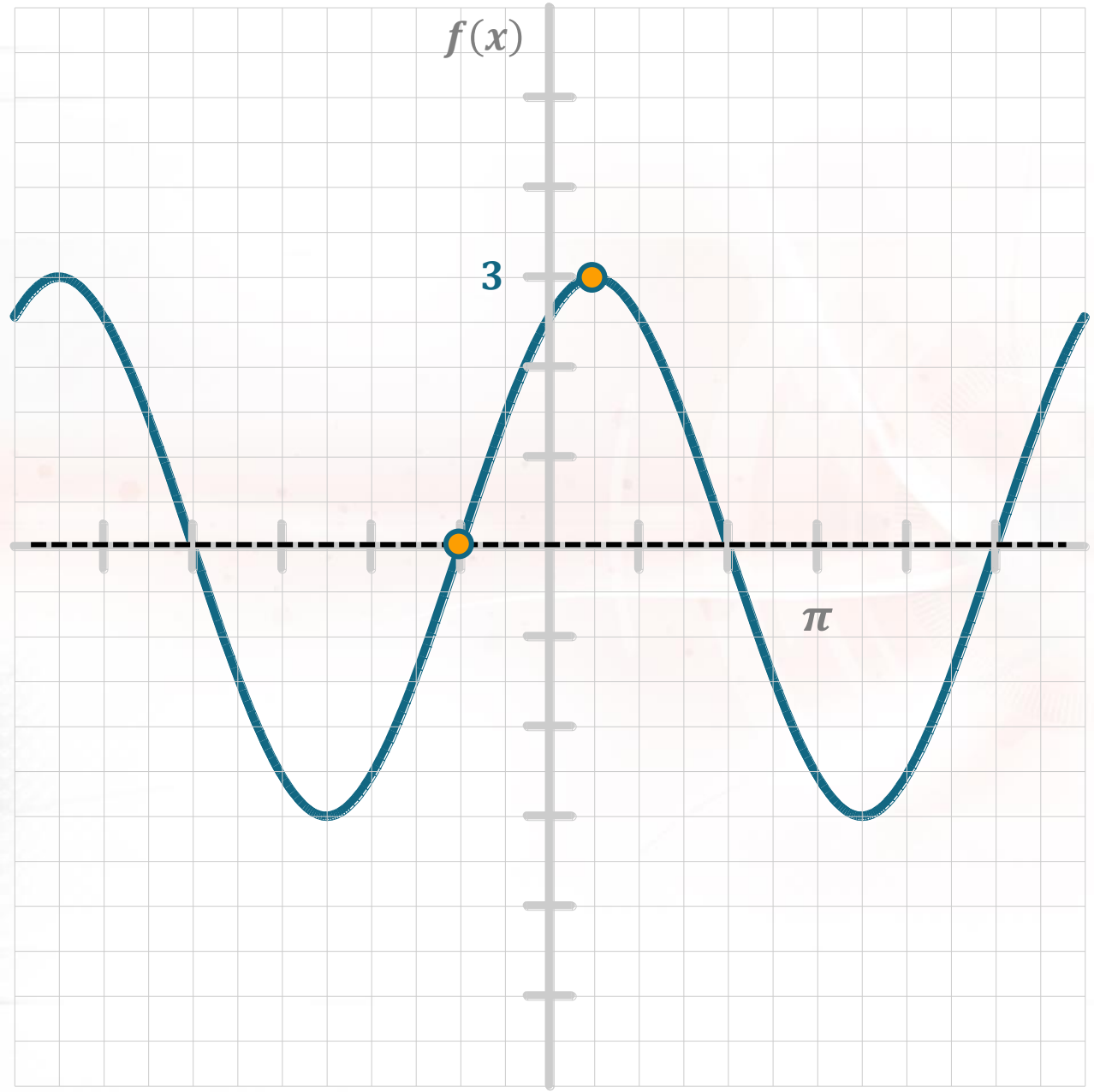

$(\sin\theta) + 1$
1
0
$\sin\theta$



The given sinusoidal function graph has its **maximum** point at $\left(\frac{\pi}{2}, 4.5\right)$ and **minimum** point at $\left(-\frac{\pi}{2}, -1.5\right)$. Find the **formula** for **f(x)**.

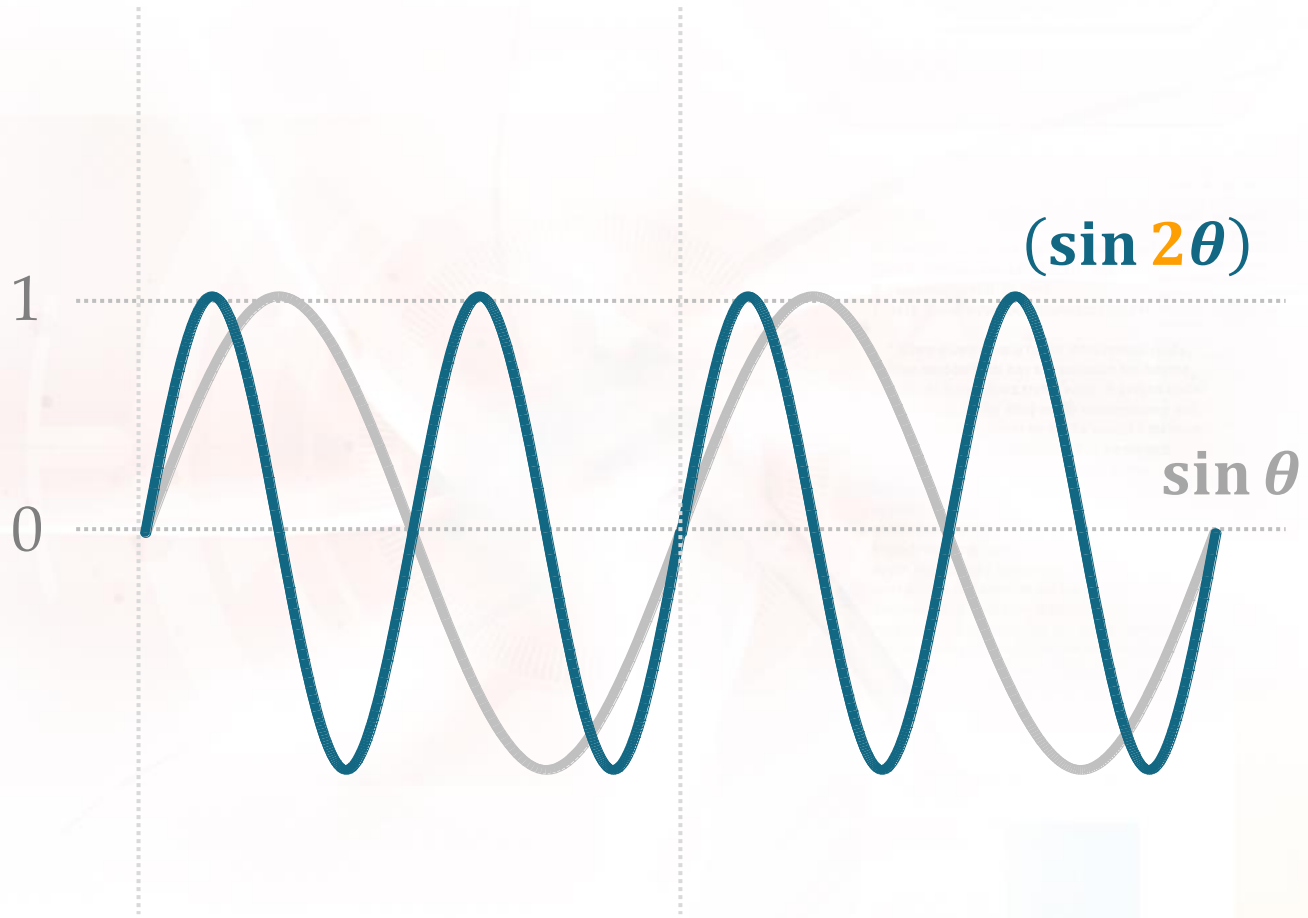
PHASE SHIFT

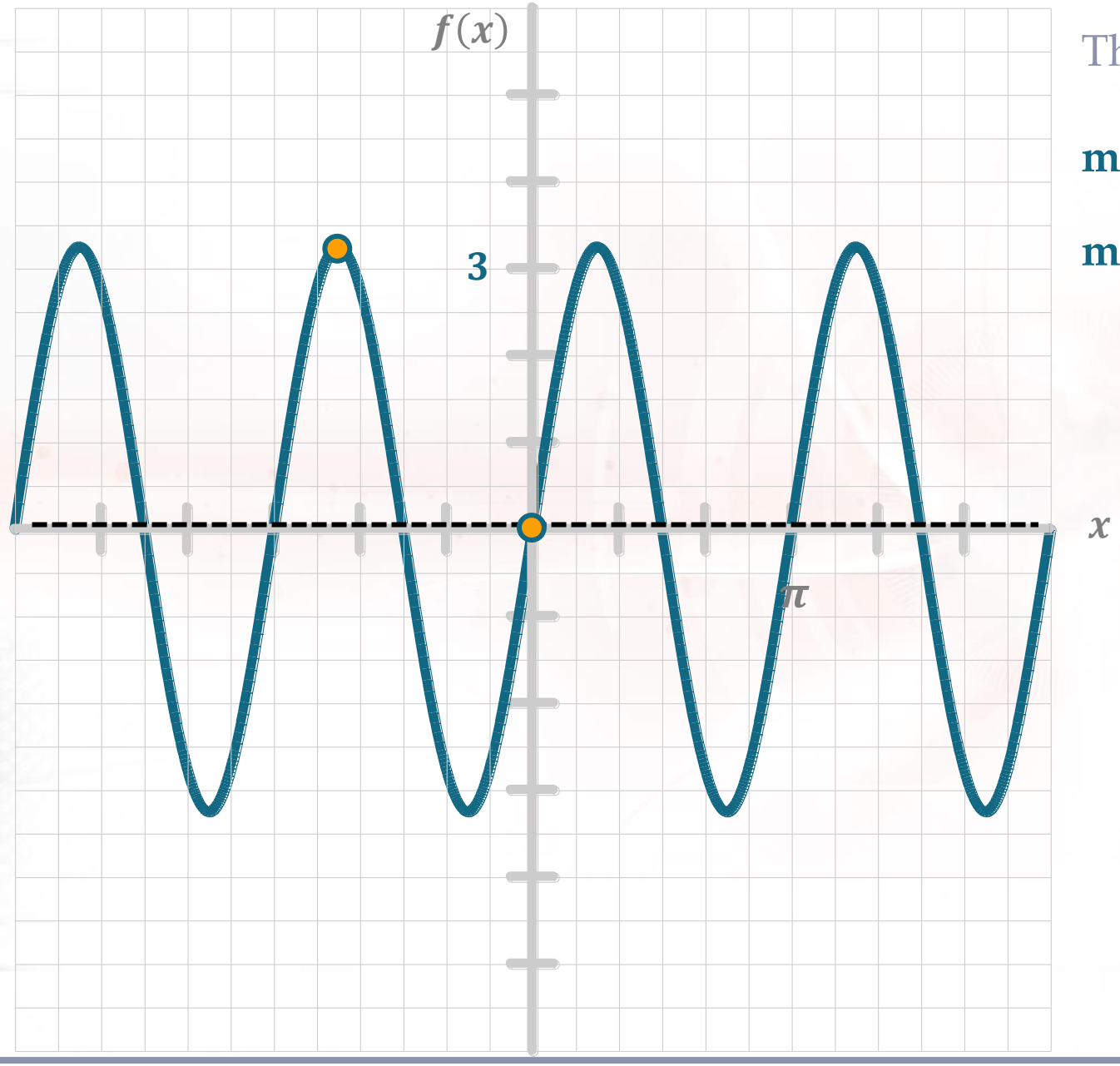




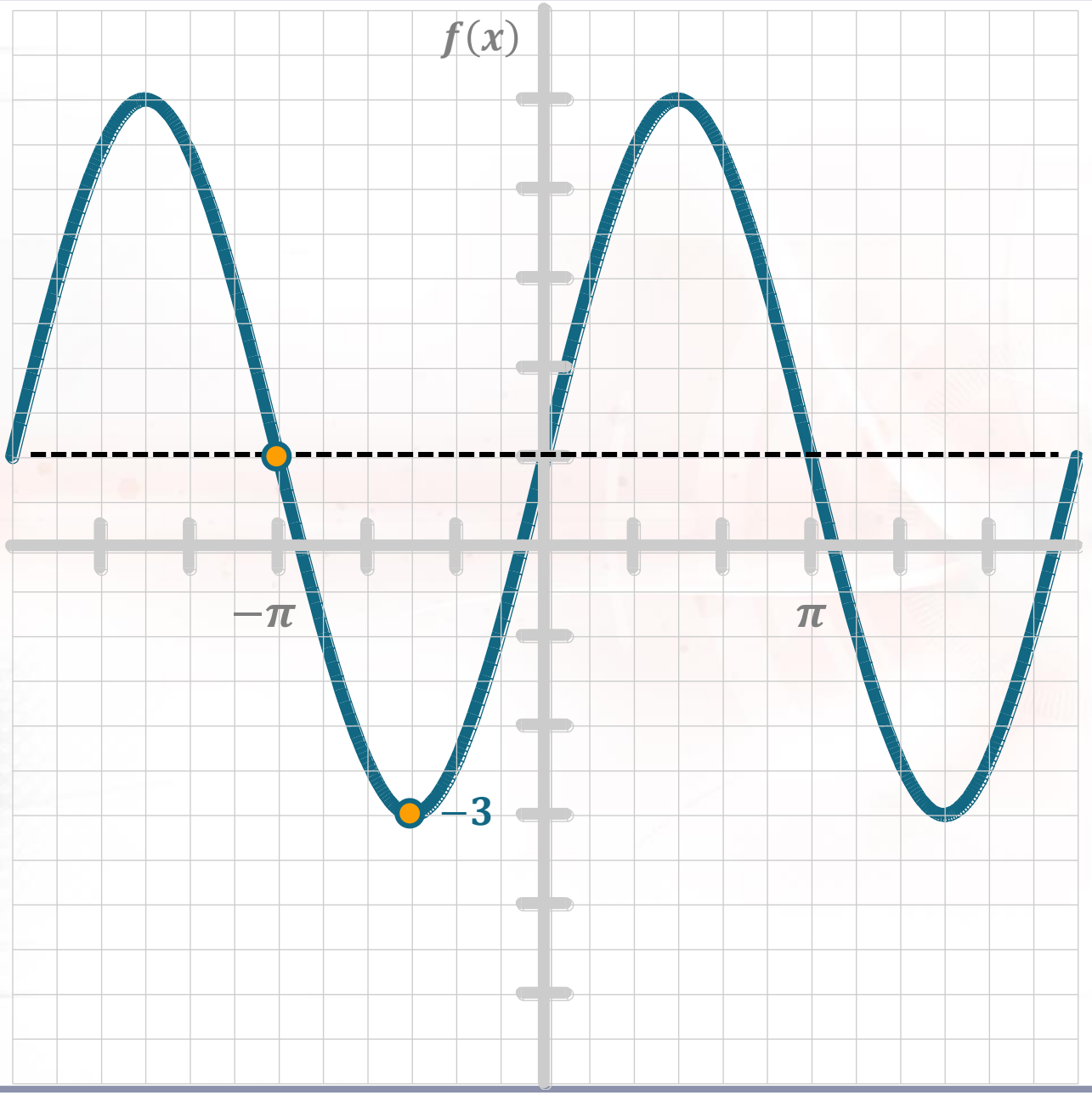
The given sinusoidal function graph has its **maximum** point at $\left(\frac{\pi}{6}, 3\right)$ and it intersects its **midline** at $\left(-\frac{\pi}{3}, 0\right)$. Find the **formula** for **f(x)**.

PERIOD

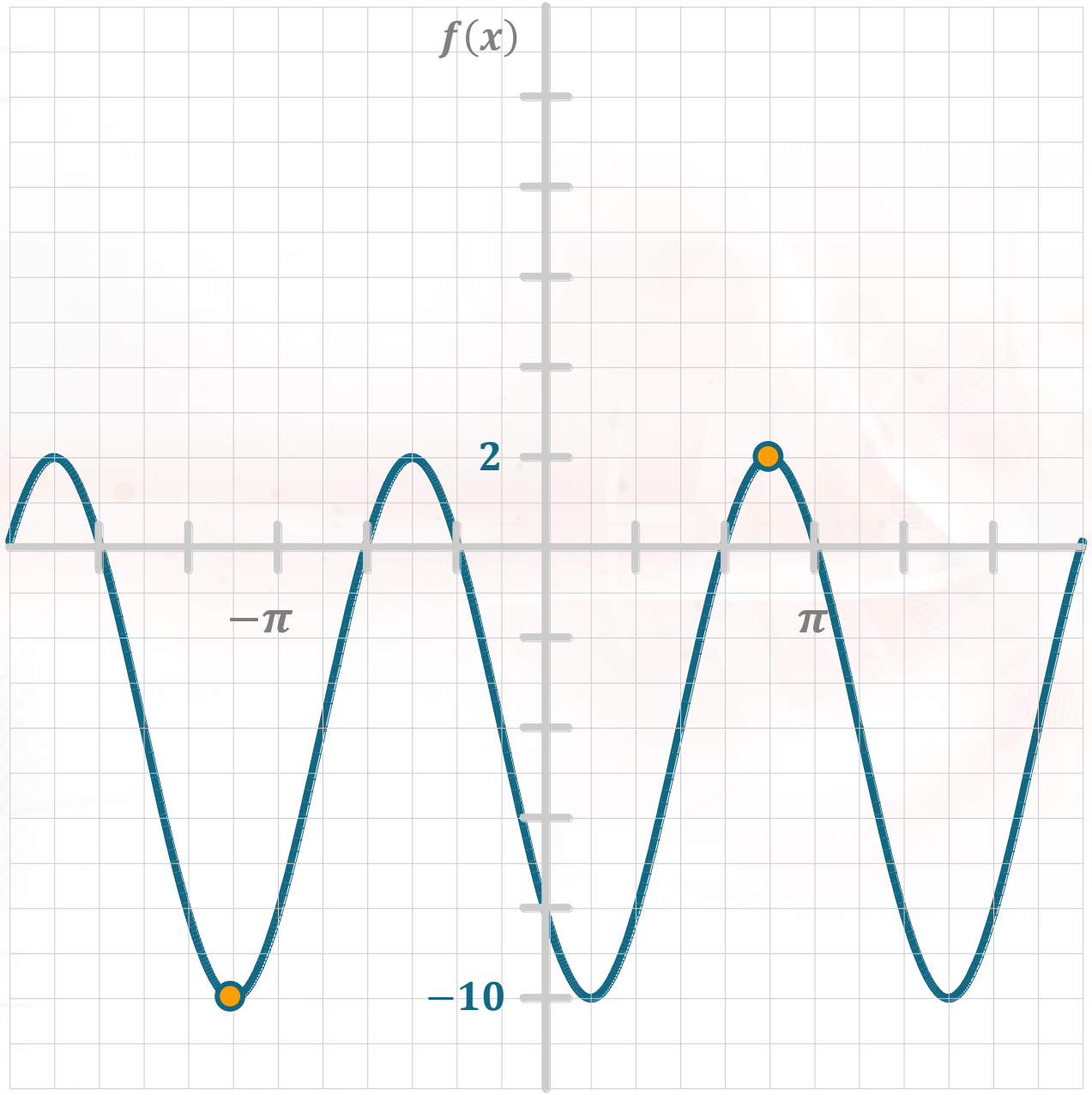




The given sinusoidal function graph has its **maximum** point at $\left(-\frac{3\pi}{4}, 3.5\right)$ and it intersects its **midline** at $(0, 0)$. Find the **formula** for **f(x)**.



The given sinusoidal function graph has its **minimum** point at $\left(-\frac{\pi}{2}, -3\right)$ and it intersects its **midline** at $(-\pi, 1)$. Find the **formula** for **f(x)**. Evaluate $f\left(\frac{\pi}{6}\right)$.



The given sinusoidal function graph has its **maximum** point at $\left(\frac{5\pi}{6}, 2\right)$ and **minimum** point at $\left(-\frac{7\pi}{6}, -10\right)$. Find the **formula** for **f(x)**. Evaluate $f\left(\frac{5\pi}{12}\right)$.

SEATWORK